Reinforcement Learning Theory

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Lecture 1: MAB and MDP Basics

Reinforcement learning studies an agent that acts on an environment, observes feedback, and seeks high cumulative reward. Actions can change both the immediate reward and the situations encountered later.

1.1 Bandits and sequential interaction

A multi-armed bandit has actions $\mathcal{A} = [A]$ and random rewards $R(a)$ with means $r(a)$. The goal is to identify an arm with largest mean. For example, $R(a) \sim \mathcal{N}(r(a), \sigma^2)$. A contextual bandit adds a context $s \in \mathcal{S}$: the learner chooses a policy, the environment reveals s, the learner samples an action from $\pi(\cdot \mid s)$, and receives its reward. A deterministic policy maps $\mathcal{S} \to \mathcal{A}$; a randomized policy maps $\mathcal{S} \to \Delta_{\mathcal{A}}$, where $\Delta_{\mathcal{A}}$ is the probability simplex over actions. The aim is a good arm for each context.

Definition 1.1 [Episodic MDP]. A Markov decision process is MDP$(\mathcal{S}, \mathcal{A}, r, P, H)$, with states $\mathcal{S}$, actions $\mathcal{A}$, horizon H, rewards $r_h(s, a)$, and transitions $P_h(s' \mid s, a)$. At step h, the agent observes s_h, chooses a_h, receives $r_h(s_h, a_h)$, and moves to $s_{h+1} \sim P_h(\cdot \mid s_h, a_h)$. Unlike contextual bandits, actions influence future states.

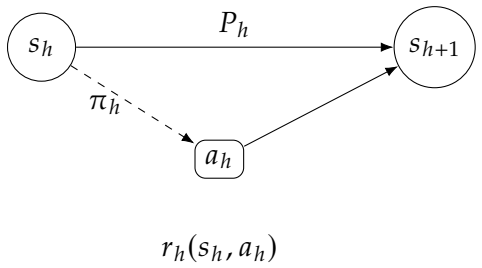


Figure 1: One step of an MDP: the policy chooses the action from the current state; the transition depends on both.

1.2 Environment, policies, and values

The environment is (P, r). In planning it is known. With a generative model, it is unknown but a simulator can answer queries (s, a, h) with a reward and sampled successor. In the interaction setting, the learner must reach states by acting from an initial state.

A Markov policy is $\pi = \{\pi_h\}_{h=1}^H$, where $\pi_h : \mathcal{S} \to \Delta_{\mathcal{A}}$. Its value functions are

$$V_h^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=h}^{H} r_t(s_t, a_t) \mid s_h = s \right],$$
$$Q_h^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=h}^{H} r_t(s_t, a_t) \mid s_h = s, a_h = a \right].$$

The goal is to maximize $V_1^\pi(s_1)$ for fixed s_1, or $\mathbb{E}_{s_1 \sim \mu} V_1^\pi(s_1)$ for an initial distribution μ.

1.3 Infinite-horizon discounting

For MDP$(\mathcal{S}, \mathcal{A}, r, P, \gamma)$ with $0 < \gamma < 1$,

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s \right].$$

If $r \in [0, 1]$, discarding rewards from time H onward loses at most $\gamma^H/(1-\gamma)$. Thus an ϵ-accurate truncation has

$$H \geq \frac{\log(1/((1-\gamma)\epsilon))}{-\log \gamma} = \widetilde{O}\left(\frac{1}{1-\gamma}\right).$$

This is the effective horizon when γ is close to one.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 1 (Spring 2020).

Lecture 2: Concentration Inequalities

2.1 Asymptotic notation

$g = O(f)$ means $g \le Cf$ eventually for a fixed constant C; $g = o(f)$ means $g/f \to 0$. The reverse relations are Ω and ω. $\widetilde{O}(f)$ suppresses logarithmic factors in the problem parameters.

2.2 Hoeffding concentration

Theorem 2.1 [Hoeffding]. *For independent $X_i \in [a_i, b_i]$, with $R_i = b_i - a_i$,*

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mathbb{E}X_i) \ge t\right) \le \exp\left(-\frac{2t^2}{\sum_i R_i^2}\right), \qquad t \ge 0.$$

In particular, if the samples are i.i.d. in an interval of length R, then with probability at least $1-\delta$,

$$\left|\frac{1}{n}\sum_i X_i - \mathbb{E}X_1\right| \le R\sqrt{\frac{\log(2/\delta)}{2n}}.$$

Proof. For any nonnegative Y, $\mathbb{P}(Y \ge u) \le \mathbb{E}Y/u$ (Markov's inequality). Apply this to $\exp(\lambda \sum_i (X_i - \mathbb{E}X_i))$ and use independence. If $\mathbb{E}e^{\lambda(X_i - \mathbb{E}X_i)} \le e^{\lambda^2\sigma_i^2/2}$, minimizing $-\lambda t + \lambda^2 \sum_i \sigma_i^2/2$ gives $\exp(-t^2/(2\sum_i \sigma_i^2))$.
For bounded X, let $\psi(\lambda) = \log \mathbb{E}e^{\lambda(X - \mathbb{E}X)}$. Then $\psi(0) = \psi'(0) = 0$ and $\psi''(\lambda)$ is the variance under the exponentially tilted law. Its support still has length R, so $\psi'' \le R^2/4$ and $\psi(\lambda) \le \lambda^2 R^2/8$. Use $\sigma_i = R_i/2$ and apply the same argument to $-X_i$ for the lower tail. □

Variables satisfying the moment-generating-function condition are sub-Gaussian; boundedness is sufficient but unnecessary. A Gaussian $\mathcal{N}(\mu, \sigma^2)$ satisfies it with parameter σ. For Bernoulli samples of mean p, Hoeffding gives error $O(\sqrt{\log(1/\delta)/n})$.

2.3 Bernstein concentration

Theorem 2.2 [Bernstein]. *If independent centered Y_i satisfy $|Y_i| \le R$ and $\mathrm{Var}(Y_i) \le \sigma_i^2$, then*

$$\mathbb{P}\left(\sum_i Y_i \ge t\right) \le \exp\left(-\frac{t^2}{2\sum_i \sigma_i^2 + 2Rt/3}\right).$$

Thus, for i.i.d. samples with variance at most σ^2, the sample-mean error is $\widetilde{O}(\sigma/\sqrt{n} + R/n)$ with high probability.

Bernstein improves on Hoeffding when variance is much smaller than squared range. For Bernoulli rewards, $\sigma^2 = p(1-p) \le p$, giving error $O(\sqrt{p\log(1/\delta)/n} + \log(1/\delta)/n)$.

2.4 Adaptive samples

In a bandit, the next arm depends on earlier rewards, so the centered rewards are generally dependent. Nevertheless, with $\mathcal{F}_{i-1}$ containing the past and the randomness used to select a_i,

$$Y_i = R_i(a_i) - r(a_i), \qquad \mathbb{E}[Y_i \mid \mathcal{F}_{i-1}] = 0.$$

This conditional centering will replace independence.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 2 (Spring 2020).

Lecture 3: Uniform Concentration and Bellman Equations

3.1 Martingale differences

A filtration $\mathcal{F}_0 \subseteq \cdots \subseteq \mathcal{F}_n$ records growing information. An adapted sequence Y_i is a martingale difference sequence if $\mathbb{E}[Y_i \mid \mathcal{F}_{i-1}] = 0$. Its partial sums form a martingale. Dependence is allowed: if X_1 is a fair bit, and X_2 is a fair sign when $X_1 = 0$ but zero when $X_1 = 1$, then $X_1 - 1/2, X_2$ are martingale differences although X_1, X_2 are dependent.

Theorem 3.1 [Azuma–Hoeffding and martingale Bernstein]. *For conditionally centered $Y_i \in [a_i, b_i]$ with deterministic interval lengths R_i,*

$$\mathbb{P}\left(\sum_i Y_i \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_i R_i^2}\right).$$

If instead $|Y_i| \leq R$ and $\mathbb{E}[Y_i^2 \mid \mathcal{F}_{i-1}] \leq \sigma_i^2$ for deterministic σ_i^2, the Bernstein bound holds with denominator $2\sum_i \sigma_i^2 + 2Rt/3$.

Proof. For Hoeffding, condition on $\mathcal{F}_{n-1}$ in the exponential moment:

$$\mathbb{E}e^{\lambda \sum_{i=1}^n Y_i} \leq e^{\lambda^2 R_n^2/8} \mathbb{E}e^{\lambda \sum_{i<n} Y_i}.$$

Iterate, then use Markov's inequality and optimize λ. The Bernstein version uses its conditional exponential-moment bound in the same iteration. Independence is never needed. □

3.2 Uniform concentration

For i.i.d. $X_i \in \mathcal{S}$ and a fixed $f : \mathcal{S} \to [0,1]$, the empirical mean concentrates at rate $\widetilde{O}(n^{-1/2})$. A data-selected $\widehat{f}$ requires a bound holding simultaneously over its function class. For finite $\mathcal{F}$, a union bound gives, with probability at least $1 - \delta$,

$$\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_i f(X_i) - \mathbb{E}f(X) \right| \leq \sqrt{\frac{\log(2|\mathcal{F}|/\delta)}{2n}}.$$

Here $\mathbb{E}f(X)$ means a fresh sample from the population law. For all functions $\mathcal{S} \to [0,1]$, where $|\mathcal{S}| = S$, discretize every coordinate on an η-grid. The grid has at most $(1 + 1/\eta)^S$ members, and replacing f by its grid approximation changes the deviation by at most 2η. Taking $\eta = n^{-1/2}$ yields $\widetilde{O}(\sqrt{S/n})$.

3.3 Bellman equations and optimal policies

Write $(P_h V)(s,a) = \mathbb{E}_{s' \sim P_h(\cdot|s,a)} V(s')$. With $V^\pi_{H+1} = 0$, conditioning on the first step gives

$$Q^\pi_h = r_h + P_h V^\pi_{h+1}, \qquad V^\pi_h(s) = \sum_a \pi_h(a \mid s) Q^\pi_h(s,a).$$

A general policy may depend on the whole history. In a fully observed MDP, a deterministic Markov policy attains the optimal value even among these general policies:

$$Q^\star_h = r_h + P_h V^\star_{h+1}, \qquad V^\star_h(s) = \max_a Q^\star_h(s,a), \qquad \pi^\star_h(s) \in \arg\max_a Q^\star_h(s,a).$$

Proof. Work backward from $H+1$. If the optimal continuation value is known at $h+1$, no randomization or additional history can improve on choosing the action maximizing $r_h + P_h V^\star_{h+1}$. This choice attains the bound at every state simultaneously. □

Evaluating a fixed policy and finding an optimal one are therefore backward dynamic programs. Policy iteration alternates evaluation and greedy improvement; after H improvements its last H decisions are optimal.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 3 (Spring 2020).

Lecture 4: MDP Planning

4.1 Finite-horizon planning

Assume $r_h \in [0,1]$. Regard V_h as an S-vector, Q_h as an SA-vector, and P_h as an $SA \times S$ row-stochastic matrix. Then $V_h \in [0, H-h+1]^S$ and $Q_h \in [0, H-h+1]^{SA}$.

Algorithm 4.1: Policy evaluation and value iteration

```
Set V_{H+1} = 0
for h = H, H-1, ..., 1 do
    Q_h(s,a) ← r_h(s,a) + (P_h V_{h+1})(s,a) for all s, a
    For evaluation: V_h(s) ← Σ_a π_h(a | s) Q_h(s,a)
    For optimization: V_h(s) ← max_a Q_h(s,a) and π_h(s) ∈ arg max_a Q_h(s,a)
```

Alternatively, start with π^0, evaluate Q^{π^t}, and set $\pi_h^{t+1}(s) \in \arg\max_a Q_h^{\pi^t}(s,a)$. After t improvements, the policy is optimal on its last t stages. Indeed, the last-stage greedy choice is optimal, and each further improvement extends the optimal suffix by one stage. Thus π^H is optimal.

4.2 Discounted policy evaluation

Let $\beta = (1-\gamma)^{-1}$. For a stationary policy,

$$Q^\pi = r + \gamma P^\pi Q^\pi, \qquad P^\pi((s',a') \mid (s,a)) = P(s' \mid s,a)\pi(a' \mid s').$$

The operator $T^\pi Q = r + \gamma P^\pi Q$ is a γ-contraction in $\|\cdot\|_\infty$, since every row of P^π sums to one. Consequently,

$$Q^{t+1} = T^\pi Q^t \quad \Longrightarrow \quad \|Q^t - Q^\pi\|_\infty \le \gamma^t \|Q^0 - Q^\pi\|_\infty.$$

For initial values in $[0,\beta]^{SA}$, $t = O(\beta \log(\beta/\epsilon))$ iterations suffice for error ϵ.

4.3 Discounted optimality and value iteration

Define $(MQ)(s) = \max_a Q(s,a)$ and $T^\star Q = r + \gamma PMQ$. Because $\|MQ - MQ'\|_\infty \le \|Q - Q'\|_\infty$, $T^\star$ is also a γ-contraction. Its unique fixed point is $Q^\star$, and value iteration $Q^{t+1} = T^\star Q^t$ has the same geometric error bound. An optimal policy is greedy with respect to $Q^\star$.

4.4 Linear programming

For strictly positive weights $\alpha(s)$, $V^\star$ uniquely solves

$$\min_{V \in [0,\beta]^S} \sum_s \alpha(s) V(s) \quad \text{subject to} \quad V(s) \geq r(s,a) + \gamma (PV)(s,a) \quad \text{for every } s, a.$$

Proof. A feasible V satisfies $V \geq TV$, where $TV = \max_a (r + \gamma PV)$. Monotonicity and contraction imply $V \geq T^t V \to V^\star$. Since $V^\star$ is feasible and all weights are positive, it is the unique minimizer. □

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 4 (Spring 2020).

Lecture 5: Generative Models and Value Iteration: Coarse Analysis

5.1 Approximation and simulator access

In a discounted MDP, let π_Q be greedy with respect to Q. Then

$$\|V^\star - V^{\pi_Q}\|_\infty \le 2(1-\gamma)^{-1}\|Q - Q^\star\|_\infty.$$

Proof. Add and subtract $Q^\star(s, \pi_Q(s))$ and the two corresponding Q values. Greediness bounds the immediate action-selection error by $2\|Q - Q^\star\|_\infty$. The continuation error is at most $\gamma\|V^\star - V^{\pi_Q}\|_\infty$. Rearrange. □

Discounting can also be reduced to finite-horizon planning by truncating after an effective horizon and using rewards $\gamma^{h-1}r$.

A simulator returns independent $s' \sim P_h(\cdot \mid s, a)$ for any requested (s, a, h). In a maze this allows teleporting to a chosen square; ordinary interaction does not. Assume deterministic known rewards, so learning the transition is the statistical challenge.

Algorithm 5.1: Model-based value iteration

1. For every (h, s, a), collect n successor samples
2. $\widehat{P}_h(s' \mid s, a) \leftarrow n^{-1}\sum_{i=1}^{n} \mathbf{1}\{s'_i = s'\}$
3. Run backward value iteration in $(\widehat{P}, r)$
4. **return** its optimal policy $\widehat{\pi}$

5.2 Simulation lemma

For any fixed policy π, let $V^\pi, \widehat{V}^\pi$ denote its values in the true and empirical models. Subtracting their Bellman equations and unrolling gives

$$V_h^\pi(s) - \widehat{V}_h^\pi(s) = \mathbb{E}_{\widehat{P},\pi}\left[\sum_{t=h}^{H} ((P_t - \widehat{P}_t)V_{t+1}^\pi)(s_t, a_t) \mid s_h = s\right].$$

The expectation uses the empirical model because the remaining value difference propagates through $\widehat{P}_t$.

5.3 Coarse analysis

Uniform concentration over all $V \in [0, H]^S$ gives, simultaneously over (h, s, a),

$$|((P_h - \widehat{P}_h)V)(s, a)| \le CH\sqrt{\frac{S\iota}{n}}, \qquad \iota = \log\frac{HSA}{\delta},$$

with logarithmic discretization factors absorbed in the bound. Equivalently, bound the transition error in ℓ_1 and multiply by $\|V\|_\infty$. The simulation lemma then gives value error at most $CH^2\sqrt{S\iota/n}$ for every policy, including data-selected policies.

Theorem 5.1 [Coarse sample bound]. *Taking $n = \widetilde{O}(H^4S/\epsilon^2)$ samples per (h, s, a) makes $\widehat{\pi}$ ϵ-optimal with high probability. Total simulator queries are $\widetilde{O}(H^5S^2A/\epsilon^2)$.*

Proof. Insert empirical-model values between $V^{\pi^\star}$ and $V^{\widehat{\pi}}$. The empirical optimality term is nonpositive. Each of the two remaining model errors is at most $\epsilon/2$ by the preceding bound. □

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 5 (Spring 2020).

Lecture 6: Value Iteration: Refined Analysis and Q-Learning

6.1 Removing a factor of the number of states

For a fixed $V \in [0, H]^S$, Hoeffding gives $|((P_h - \widehat{P}_h)V)(s, a)| \le CH\sqrt{\iota/n}$. This also holds for random V independent of the samples used in $\widehat{P}_h(\cdot \mid s, a)$, by conditioning on V. Uniform control over every data-adapted V costs an extra $\sqrt{S}$.

Theorem 6.1 [Refined value-iteration bound]. *With independent samples at different stages, $n = \widetilde{O}(H^4/\epsilon^2)$ per (h, s, a) suffices for an ϵ-optimal policy, for total cost $\widetilde{O}(H^5SA/\epsilon^2)$.*

Proof. Use the same three-term decomposition as before. For $\pi^\star$, its true continuation values are fixed. For $\widehat{\pi}$, its true value at $h + 1$ depends only on empirical transitions from stages $h+1, \dots, H$, so it is independent of the samples at stage h. Apply fixed-function concentration to these two continuation values, then the simulation lemma. Each model error is $\widetilde{O}(H^2/\sqrt{n})$. □

6.2 Q-learning

Q-learning estimates values directly rather than storing an empirical model. In synchronous rounds, sample one successor s'_t per (h, s, a) and update

$$Q_h^t(s, a) = (1 - \alpha_t)Q_h^{t-1}(s, a) + \alpha_t\big(r_h(s, a) + V_{h+1}^{t-1}(s'_t)\big), \qquad V_h^t(s) = \max_a Q_h^t(s, a).$$

Initialize $Q_h^0 = 0$ and $V_{H+1}^t = 0$. Only $O(HSA)$ values need storing. For $\alpha_1 = 1$, unrolling the update gives weights

$$\gamma_t^i = \alpha_i \prod_{j=i+1}^{t} (1 - \alpha_j), \qquad \sum_{i=1}^{t} \gamma_t^i = 1.$$

Choosing $\alpha_t = 1/t$ gives uniform weights; the choice $\alpha_t = (H + 1)/(H + t)$ favors recent, more accurate targets.

Lemma 6.2 [Learning-rate weights]. *For this choice,*

$$\sum_{i=1}^{t} (\gamma_t^i)^2 \le \frac{2H}{t}, \qquad \sum_{t=i}^{\infty} \gamma_t^i \le 1 + \frac{1}{H}.$$

6.3 Error recursion

Subtract $Q_h^\star = r_h + P_h V_{h+1}^\star$ from the unrolled update. Weighted Hoeffding bounds the noise from $V_{h+1}^\star(s_i')$ by $O(\sqrt{H^3\iota/t})$. The remaining error satisfies

$$e_h(t) \le \sum_{i=1}^{t} \gamma_t^i e_{h+1}(i-1) + C\sqrt{H^3\iota/t}, \qquad e_h(t) = \|Q_h^t - Q_h^\star\|_\infty.$$

Here ι includes the logarithm needed for a union bound over the chosen finite number of rounds. The weight-sum lemma prevents an exponential loss when this recursion is propagated across stages.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 6 (Spring 2020).

Lecture 7: Generative Model Summary and Multi-Armed Bandits

7.1 Finishing the Q-learning bound

Sum the preceding error recursion over rounds, interchange the two sums, and use $\sum_{j=i}^{t} \gamma_j^i \le 1+1/H$ and $\sum_{j=1}^{t} j^{-1/2} \le 2\sqrt{t}$. Including the initial error in the constant,

$$\sum_{j=0}^{t} e_h(j) \le \left(1 + \frac{1}{H}\right) \sum_{j=0}^{t} e_{h+1}(j) + C\sqrt{H^3 t\iota}.$$

Unrolling over H stages costs at most $\sum_{h=1}^{H}(1+1/H)^h \le eH$. Consequently,

$$\left\| \frac{1}{t} \sum_{i=1}^{t} Q_1^i - Q_1^\star \right\|_\infty \le \frac{1}{t} \sum_{i=1}^{t} e_1(i) \le C\sqrt{\frac{H^5\iota}{t}}.$$

The same argument applies to every stage. Thus estimating the Q-functions to accuracy ϵ takes $\widetilde{O}(H^6SA/\epsilon^2)$ queries. This is a Q-estimation guarantee; converting a uniformly approximate Q-function to a greedy policy can incur an additional horizon factor.

7.2 Generative-model comparison

The preceding value-iteration policy bound is $\widetilde{O}(H^5SA/\epsilon^2)$; sharper variance-sensitive analysis improves it to $\widetilde{O}(H^4SA/\epsilon^2)$ for nonstationary episodic MDPs, matching the minimax lower bound. Stationary transitions allow the rate $\widetilde{O}(H^3SA/\epsilon^2)$. A model requires up to $O(HS^2A)$ storage, whereas Q-learning stores $O(HSA)$ entries.

7.3 Exploration versus exploitation

Without a simulator, learning a useful state may first require learning how to reach it. For a bandit with rewards in $[0,1]$ and best arm $a^\star$,

$$\text{Regret}(T) = \sum_{t=1}^{T} \big(r(a^\star) - r(a_t)\big).$$

Best-arm identification asks only for a good final recommendation; low regret also requires good decisions while learning.

If regret is at most $cT^{1-\alpha}$, choosing a uniformly random past action has average suboptimality at

most $cT^{-\alpha}$, so $T \geq (c/\epsilon)^{1/\alpha}$ suffices. Conversely, an identification procedure using $c\epsilon^{-\beta}$ pulls can be followed by repeatedly playing its recommendation:

$$\mathrm{Regret}(T) \leq c\epsilon^{-\beta} + T\epsilon = O\big(c^{1/(1+\beta)}T^{\beta/(1+\beta)}\big)$$

after balancing the terms. These conversions use the same expected or high-probability convention as the assumed guarantees.

7.4 Simple exploration rules

Explore then commit samples each arm n times, then plays the arm with largest empirical mean. Hoeffding and a union bound show that $n = O(\log(A/\delta)/\epsilon^2)$ identifies an ϵ-optimal arm. Balancing exploration and exploitation gives $\widetilde{O}(A^{1/3}T^{2/3})$ regret.

ϵ-greedy continually updates empirical means, choosing a uniform arm with probability ϵ and an empirical best arm otherwise. A horizon-dependent exploration rate, balancing the random-pull cost with estimation error, gives the same $\widetilde{O}(A^{1/3}T^{2/3})$ order.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 7 (Spring 2020).

Lecture 8: UCB and Exploration in MDPs

8.1 Optimism for bandits

Let $N_t(a)$ count pulls before round t, and let $\widehat{r}_t(a)$ be the empirical mean. Set $\iota = \log(AT/\delta)$ and

$$U_t(a) = \begin{cases} \widehat{r}_t(a) + c\sqrt{\iota / N_t(a)}, & N_t(a) > 0, \\ 1, & N_t(a) = 0. \end{cases}$$

The upper confidence bound (UCB) rule picks an arm maximizing U_t. Uncertainty raises an arm's index and induces exploration.

Theorem 8.1 [UCB regret]. *With probability at least* $1 - \delta$, $\mathrm{Regret}(T) = O(\sqrt{AT\iota})$ *for* $T \geq A$.

Proof. Concentration simultaneously over each arm and each possible visit count gives $r(a) \leq U_t(a)$ and $U_t(a) - r(a) \leq C\sqrt{\iota / \max\{1, N_t(a)\}}$. The adaptive count causes no difficulty because the bound holds for every prefix of that arm's reward stream. Optimism and the greedy choice give

$$r(a^\star) - r(a_t) \leq U_t(a_t) - r(a_t).$$

Regroup by arm and visit count. Since $\sum_{j=1}^{n} j^{-1/2} \leq 2\sqrt{n}$ and $\sum_a N_{T+1}(a) = T$, the total is at most $C\sqrt{\iota} \sum_a \sqrt{N_{T+1}(a)} \leq C\sqrt{AT\iota}$. □

Explore-then-commit and ϵ-greedy have $T^{2/3}$ regret; UCB attains the minimax $\sqrt{T}$ order up to logarithms. If $T < A$, linear regret is unavoidable in general.

8.2 Contextual bandits

For S contexts, run a separate UCB instance in each context. If context s appears T_s times, its regret is $\widetilde{O}(\sqrt{AT_s})$. Cauchy–Schwarz and $\sum_s T_s = T$ give

$$\mathrm{Regret}(T) \leq C\sqrt{\iota} \sum_s \sqrt{AT_s} \leq C\sqrt{SAT\iota}.$$

8.3 Why random exploration fails in MDPs

A two-state COMBINATION LOCK has a good state g and an absorbing bad state b at each stage. One action at g preserves g; another moves to b. Only reaching g at the last stage earns reward one. Uniform random actions survive $H-1$ transitions with probability $2^{-(H-1)}$, requiring $\Omega(2^H)$ episodes to see a reward.

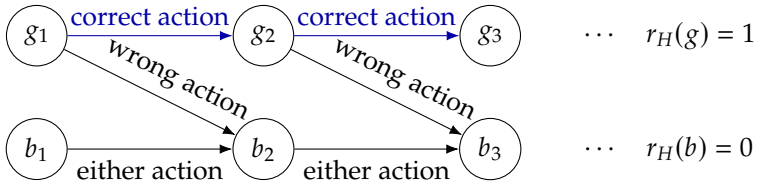


Figure 2: The combination lock. State observations reveal whether a chosen action stayed on the good path, even before a terminal reward is seen.

This is a failure of random exploration, rather than a lower bound for all algorithms: an optimistic learner can use the observed transitions to explore the good path systematically.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 8 (Spring 2020).

Lecture 9: UCB Value Iteration

9.1 The UCB-VI algorithm

Let K be the number of episodes, $T = KH$ the number of transitions, and $\iota = \log(HSAT/\delta)$. Before episode k, estimate transitions from previous episodes and retain counts $N_h^k(s,a)$.

Algorithm 9.1: UCB value iteration

Initialize counts to zero
for episode $k = 1, \dots, K$ **do**
  Set $V_{H+1}^k = 0$
  for $h = H, H-1, \dots, 1$ **do**
    For unvisited (s,a), set $Q_h^k(s,a) = H$
    Otherwise set $Q_h^k(s,a) \leftarrow \min\{H, r_h(s,a) + (\widehat{P}_h^k V_{h+1}^k)(s,a) + cH\sqrt{\iota / N_h^k(s,a)}\}$
    Set $V_h^k(s) \leftarrow \max_a Q_h^k(s,a)$
  Execute $\pi_h^k(s) \in \arg\max_a Q_h^k(s,a)$ and update the counts

Theorem 9.1 [UCB-VI regret]. *With probability at least* $1-\delta$,

$$\mathrm{Regret}(K) := \sum_{k=1}^{K} \left(V_1^\star(s_1) - V_1^{\pi^k}(s_1)\right) \le C\left(\sqrt{H^3SAT\iota} + H^3S^2A\iota^2\right).$$

9.2 Optimism

Backward induction gives $Q_h^k \ge Q_h^\star$ and $V_h^k \ge V_h^\star$. For a visited pair, subtract $Q_h^\star$ before clipping:

$$\widehat{P}_h^k(V_{h+1}^k - V_{h+1}^\star) + (\widehat{P}_h^k - P_h)V_{h+1}^\star + b_h^k.$$

The first term is nonnegative by induction; the bonus dominates the second by fixed-function concentration. Clipping at H preserves optimism.

9.3 Regret recursion

Let $\Delta_h^k = V_h^k - V_h^{\pi^k} \ge 0$. Along the executed trajectory, greediness and Bellman evaluation give

$$\Delta_h^k(s_h^k) \le P_h\Delta_{h+1}^k(s_h^k, a_h^k) + (\widehat{P}_h^k - P_h)(V_{h+1}^k - V_{h+1}^\star)(s_h^k, a_h^k) + 2b_h^k.$$

Entrywise Bernstein concentration, applied to the nonnegative function $G = V_{h+1}^k - V_{h+1}^\star$, bounds the middle term by

$$\frac{1}{H}(P_h G)(s_h^k, a_h^k) + \frac{CH^2 S\iota}{\max\{1, N_h^k(s_h^k, a_h^k)\}}.$$

Use $G \le \Delta_{h+1}^k$, and replace $P_h \Delta_{h+1}^k$ by its realized successor value plus a centered martingale difference ξ_h^k. This yields a recursion with multiplier $1 + 1/H$ and remainder $\xi_h^k + \zeta_h^k + 2b_h^k$, where $\zeta_h^k = CH^2 S\iota / \max\{1, N_h^k\}$. Unrolling introduces weights at most $(1 + 1/H)^H \le e$.

9.4 Counting the bonuses

Regrouping by (h, s, a) and visit number gives

$$\sum_{k,h} b_h^k \le C\sqrt{H^3 SAT\iota}, \qquad \sum_{k,h} \zeta_h^k \le CH^3 S^2 A\iota \log T.$$

The first uses $\sum_{j \le n} j^{-1/2} \le 2\sqrt{n}$ and Cauchy–Schwarz; the second uses the harmonic-sum bound. The martingale term is handled next.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 9 (Spring 2020).

Lecture 10: Q-Learning with UCB and MDP Summary

10.1 Completing the UCB-VI proof

The error

$$\xi_h^k = (P_h \Delta_{h+1}^k)(s_h^k, a_h^k) - \Delta_{h+1}^k(s_{h+1}^k)$$

has conditional mean zero given the history before the transition, because Δ^k was fixed before the episode. Its magnitude is $O(H)$. The deterministic stage weights from unrolling are at most e, so Azuma–Hoeffding bounds their weighted sum by $O(H\sqrt{T\iota})$. Together with the bonus sums, this proves

$$\text{Regret}(K) \le C\left(\sqrt{H^3SAT\iota} + H^3S^2A\iota^2\right).$$

Uniformly selecting one of the K episode policies gives value gap $\text{Regret}(K)/K$. Thus

$$T = \widetilde{O}\left(\frac{H^5SA}{\epsilon^2} + \frac{H^4S^2A}{\epsilon}\right)$$

suffices for an ϵ-optimal mixture at the initial state. The second term is a burn-in cost in this analysis.

In the combination lock, unvisited actions have high optimistic values. Observed transitions reduce the value of wrong branches, directing the learner toward unexplored parts of the good path.

10.2 Q-learning with UCB

Algorithm 10.1: Optimistic Q-learning

Initialize $Q_h(s,a) = H$, $N_h(s,a) = 0$, and $V_h(s) = H$ for $h \le H$
Set $V_{H+1} = 0$
for each episode and each step h **do**
  Choose $a_h \in \arg\max_a Q_h(s_h, a)$ and observe s_{h+1}
  $t \leftarrow N_h(s_h, a_h) + 1$; update the count
  $\alpha_t \leftarrow (H+1)/(H+t)$; $b_t \leftarrow c\sqrt{H^3\iota/t}$
  $Q_h(s_h, a_h) \leftarrow (1-\alpha_t)Q_h(s_h, a_h) + \alpha_t(r_h(s_h, a_h) + V_{h+1}(s_{h+1}) + b_t)$
  $V_h(s_h) \leftarrow \min\{H, \max_a Q_h(s_h, a)\}$

Its high-probability regret is $\widetilde{O}(\sqrt{H^4SAT})$. The proof first establishes optimism using the learning-rate weights, then propagates and sums the resulting errors. The algorithm is model-free: it stores $O(HSA)$ values and counts and performs one local update per step.

10.3 Rates and learning settings

For nonstationary transitions, the leading rates are

Method	Leading regret, logarithms suppressed
UCB-VI, Hoeffding	$\sqrt{H^3SAT}$
UCB-VI, Bernstein	$\sqrt{H^2SAT}$
Q-learning, Hoeffding	$\sqrt{H^4SAT}$
Q-learning, Bernstein	$\sqrt{H^3SAT}$
Minimax lower bound	$\sqrt{H^2SAT}$

The UCB-VI bounds and the listed Bernstein Q-learning analysis also contain lower-order terms. Model-based planning costs more than local Q updates; replanning at suitable count thresholds can reduce repeated planning work.

With a simulator, uniform accuracy over all initial states is possible. With interaction, the objective concerns the actual initial-state distribution: very unlikely states may remain poorly estimated while contributing little to the value gap. Learning every transition accurately is unnecessary.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 10 (Spring 2020).

Lecture 11: Lower Bounds for Multi-Armed Bandits

11.1 What a lower bound asserts

An upper bound exhibits an algorithm that works for every instance. A minimax lower bound states that every algorithm has a hard instance. For bandits with $T \geq A \geq 2$ and rewards in $[0, 1]$, the target is

$$\inf_{\mathcal{A}} \sup_{M} \mathbb{E}_M \operatorname{Regret}_{\mathcal{A}}(T) \geq c\sqrt{AT}.$$

11.2 Two-arm intuition

Consider Bernoulli means $1/2 + \epsilon$ and $1/2$. Empirical means based on n pulls fluctuate on the scale $n^{-1/2}$, so distinguishing the arms requires order ϵ^{-2} samples. Taking ϵ of order $T^{-1/2}$ gives regret of order $\sqrt{T}$. This explains the scale, but analyzing one sampling rule alone does not prove a bound for all adaptive algorithms.

11.3 A distribution of hard instances

Choose a special arm a uniformly from $[A]$. Its reward mean is $1/2 + \epsilon$; every other arm has mean $1/2$. Let $\mathbb{P}_a$ be the distribution of the complete observed history under this instance, and $\mathbb{P}_0$ the distribution when every mean is $1/2$. If N_a counts pulls of arm a, then

$$\mathbb{E}_D \operatorname{Regret}(T) = \epsilon \left(T - \frac{1}{A} \sum_{a=1}^{A} \mathbb{E}_a N_a \right).$$

It suffices to show that an algorithm cannot concentrate most pulls on the randomly chosen special arm.

Note. Fixing a randomized algorithm's internal random seed makes it deterministic. If one instance distribution is hard for every deterministic algorithm, averaging over the seed gives the same lower bound for randomized algorithms. A worst instance is at least as hard as the average instance under this distribution.

11.4 Comparing history distributions

For a fixed deterministic algorithm, actions are functions of earlier observations, so N_a is a function of the reward history and lies in $[0, T]$. Total variation therefore gives

$$\mathbb{E}_a N_a - \mathbb{E}_0 N_a \le T \|\mathbb{P}_a - \mathbb{P}_0\|_{\mathrm{TV}} \le T\sqrt{\tfrac{1}{2}\,\mathrm{KL}(\mathbb{P}_0 \| \mathbb{P}_a)},$$

where the last step is Pinsker's inequality. The adaptive-history KL divergence is tractable because the chain rule separates it into the reward divergences of the arms actually pulled. The KL calculation in Lecture 12 completes the bound and transfers the construction to MDPs.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 11 (Spring 2020).

Lecture 12: Lower Bounds for MDPs

12.1 Finishing the bandit lower bound

In the hard instances above, the history KL divergence is

$$\mathrm{KL}(\mathbb{P}_0 \| \mathbb{P}_a) = \mathbb{E}_0 N_a \ \mathrm{KL}(\mathrm{Ber}(1/2) \| \mathrm{Ber}(1/2 + \epsilon)) = \frac{\mathbb{E}_0 N_a}{2} \log \frac{1}{1 - 4\epsilon^2}.$$

Proof. Factor the history probability into conditional probabilities at each round. The action rule is identical in both instances. Only observations obtained when arm a is selected contribute a nonzero conditional KL divergence. Summing their expected number under $\mathbb{P}_0$ gives the formula. □

Since $\sum_a \mathbb{E}_0 N_a = T$, Pinsker and Cauchy–Schwarz imply

$$\frac{1}{A} \sum_a \mathbb{E}_a N_a \le \frac{T}{A} + T \sqrt{\frac{T}{4A} \log \frac{1}{1 - 4\epsilon^2}}.$$

Take $\epsilon = c \min\{1, \sqrt{A/T}\}$ for a sufficiently small constant. The bracket in the regret formula stays bounded away from zero, yielding

$$\inf_{\mathcal{A}} \sup_M \mathbb{E}_M \mathrm{Regret}(T) = \Omega\big(\min\{T, \sqrt{AT}\}\big).$$

In particular this matches UCB up to logarithms for $T \ge A$. Related testing arguments show that an algorithm using T pulls and returning an arm ψ has worst-case expected suboptimality $\Omega(\min\{1, \sqrt{A/T}\})$. Thus constant-confidence ϵ-optimal identification needs $\Omega(A/\epsilon^2)$ samples.

12.2 MDP rates

Writing $T = KH$, the minimax leading regret is $\Omega(\sqrt{H^2SAT})$ for nonstationary episodic MDPs and $\Omega(\sqrt{HSAT})$ for stationary transitions, in the parameter ranges where these are below the trivial reward bound. Bernstein UCB-VI matches these leading orders. The construction below proves the stationary rate.

12.3 A hard stationary MDP

There is a decision state x and two absorbing states, g and b. At x, one unknown special action reaches g with probability $1/2 + \epsilon$; other actions do so with probability $1/2$. The remaining

probability goes to b. Each later step rewards one at g and zero at b.

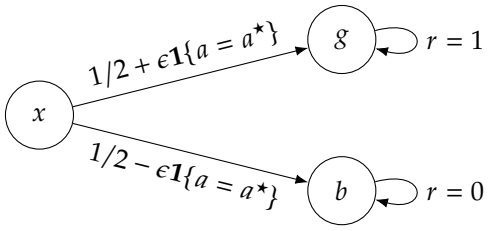


Figure 3: Only the first transition reveals information about the special action; its effect on reward persists for $H - 1$ steps.

Every episode supplies one bandit observation, but multiplies its reward gap by $H - 1$. Hence the bandit lower bound gives $\Omega(H\sqrt{AK}) = \Omega(\sqrt{HAT})$ regret. Use $m = \lfloor S/3 \rfloor$ disjoint copies with independent special actions and cycle the initial state among them. Each receives approximately K/m episodes, giving

$$m\,\Omega\left(H\sqrt{AK/m}\right) = \Omega(\sqrt{HSAT}).$$

The construction permits the indicated sequence of initial states and uses time-independent transitions.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 12 (Spring 2020).

Lecture 13: Policy Gradient for Multi-Armed Bandits

13.1 Policy optimization and mirror ascent

Parameterize policies by π_θ and maximize $J(\theta) = V_1^{\pi_\theta}(s_1)$. Projected gradient ascent solves

$$\theta^{t+1} = \arg\max_{\theta \in \Theta} \left\{ \langle \nabla J(\theta^t), \theta - \theta^t \rangle - \frac{\|\theta - \theta^t\|_2^2}{2\eta} \right\}.$$

Mirror ascent replaces the quadratic penalty by $D_\Phi(\theta, \theta^t)/\eta$, a Bregman divergence. For a bandit, $\theta \in \Delta_{\mathcal{A}}$, $J(\theta) = \langle r, \theta \rangle$, and KL regularization gives

$$\theta^{t+1}(a) = \frac{\theta^t(a) e^{\eta r(a)}}{\sum_b \theta^t(b) e^{\eta r(b)}}.$$

This follows by differentiating the Lagrangian with a multiplier enforcing $\sum_a \theta(a) = 1$. With positive initial probabilities the solution is positive, so the nonnegativity constraints are satisfied.

13.2 Known bandit rewards

For rewards in $[0, 1]$ and uniform θ^1,

$$\sum_{t=1}^{T} \langle r, \theta^\star - \theta^t \rangle \le 1 + \frac{\log A}{\eta}.$$

Proof. The update identity and cancellation of the normalization constant give

$$\begin{aligned} \eta \langle r, \theta^{t+1} - \theta \rangle &= \mathrm{KL}(\theta^{t+1} \| \theta^t) + \mathrm{KL}(\theta \| \theta^{t+1}) - \mathrm{KL}(\theta \| \theta^t) \\ &\ge \mathrm{KL}(\theta \| \theta^{t+1}) - \mathrm{KL}(\theta \| \theta^t). \end{aligned}$$

Rearrange with $\theta = \theta^\star$. Summing telescopes both the KL terms and $\langle r, \theta^{t+1} - \theta^t \rangle$, bounded by one in total. The initial KL is at most $\log A$. □

Thus the average iterate has $O(1/T)$ suboptimality for fixed η. As $\eta \to \infty$, one update concentrates on the best-reward actions.

13.3 Unknown rewards and EXP3

Only $R_t(a_t)$ is observed, with $a_t \sim \theta^t$. Define

$$\widehat{\ell}_t(a) = \frac{1 - R_t(a_t)}{\theta^t(a)} \mathbf{1}\{a = a_t\}, \qquad \widehat{r}_t(a) = 1 - \widehat{\ell}_t(a).$$

These are unbiased loss and reward estimates conditional on the past. The reward estimate need not lie in $[0, 1]$; the loss estimate is nonnegative.

Algorithm 13.1: EXP3

Initialize θ^1 uniformly
for $t = 1, \dots, T$ **do**
    Sample $a_t \sim \theta^t$ and observe $R_t(a_t)$
    Form $\widehat{\ell}_t$ by importance weighting
    $\theta^{t+1}(a) \leftarrow \theta^t(a) e^{-\eta \widehat{\ell}_t(a)} / Z_t$ for every a

The equivalent reward-form update uses $e^{\eta \widehat{r}_t(a)}$; its common factor e^{η} cancels. With $\eta = \sqrt{\log A / (AT)}$, EXP3 has expected regret $O(\sqrt{AT \log A})$. Unlike ordinary stochastic UCB, it also handles nonanticipating adversarial reward sequences, comparing with any fixed arm.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 13 (Spring 2020).

Lecture 14: Policy Gradient for MDPs

14.1 Finishing the EXP3 argument

The mirror-ascent inequality gives

$$\langle \widehat{r}_t, \theta^\star - \theta^t \rangle \le \langle \widehat{r}_t, \theta^{t+1} - \theta^t \rangle + \frac{\mathrm{KL}(\theta^\star \| \theta^t) - \mathrm{KL}(\theta^\star \| \theta^{t+1})}{\eta}.$$

The first term cannot be replaced directly by its mean because θ^{t+1} depends on the observed reward. Instead use the nonnegative loss estimate and its update normalization $Z_t \le 1$:

$$\langle \widehat{r}_t, \theta^{t+1} - \theta^t \rangle = \sum_a \widehat{\ell}_t(a)(\theta^t(a) - \theta^{t+1}(a)) \le \eta \sum_a \theta^t(a)\widehat{\ell}_t(a)^2.$$

Here $e^{-x} \ge 1 - x$ supplies the last inequality. Conditional expectation of the last sum is at most A, since losses lie in $[0, 1]$. Unbiasedness and telescoping give

$$\mathbb{E}\,\mathrm{Regret}(T) \le \eta AT + \frac{\log A}{\eta} \le 2\sqrt{AT \log A}.$$

14.2 The policy gradient in a tabular MDP

For known P, r, parameterize the policy directly by probabilities $\pi_h(a \mid s)$ and let $d_h^\pi(s) = \mathbb{P}_\pi(s_h = s)$. Holding other stages fixed, the objective is linear in the probabilities at stage h. Therefore

$$\frac{\partial J(\pi)}{\partial \pi_h(a \mid s)} = d_h^\pi(s) Q_h^\pi(s, a), \qquad \langle \nabla J(\pi), \widetilde{\pi} - \pi \rangle = \mathbb{E}_\pi \sum_{h=1}^{H} \langle Q_h^\pi(s_h, \cdot), \widetilde{\pi}_h - \pi_h \rangle.$$

The joint objective is multilinear rather than generally concave.

14.3 Occupancy-weighted policy mirror ascent

Penalize changes using KL weighted by the current occupancy:

$$\pi^{t+1} = \arg\max_\pi \left\{ \langle \nabla J(\pi^t), \pi - \pi^t \rangle - \frac{1}{\eta} \sum_{h,s} d_h^{\pi^t}(s)\, \mathrm{KL}(\pi_h(\cdot \mid s) \| \pi_h^t(\cdot \mid s)) \right\}.$$

At each state with positive occupancy the weight cancels, giving

$$\pi_h^{t+1}(a \mid s) \propto \pi_h^t(a \mid s) \exp\bigl(\eta Q_h^{\pi^t}(s,a)\bigr).$$

Use this same update at all states, including zero-occupancy ones. Initialize uniformly and evaluate Q^{π^t} by backward recursion each round. As $\eta \to \infty$ the update becomes greedy policy improvement, with ties retaining the old relative probabilities. Its cumulative suboptimality obeys

$$\sum_{t=1}^{T} \bigl(V_1^\star(s_1) - V_1^{\pi^t}(s_1)\bigr) \le H^2 + \frac{H \log A}{\eta}.$$

The relation between this mirror-ascent rule and natural policy gradient requires specifying the parameterization.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 14 (Spring 2020).

Lecture 15: PPO, TRPO, and Policy Mirror Ascent

15.1 Performance difference

Lemma 15.1 [Performance difference]. *For two Markov policies π, π',*

$$V_1^{\pi}(s_1) - V_1^{\pi'}(s_1) = \mathbb{E}_{\pi} \sum_{h=1}^{H} \langle Q_h^{\pi'}(s_h, \cdot), \pi_h(\cdot \mid s_h) - \pi'_h(\cdot \mid s_h) \rangle.$$

Proof. Interpolate between the policies by using π for the first h steps and π' afterward. Successive hybrid policies differ only at step h; their difference is the displayed inner product under the π roll-in. Sum these differences. □

15.2 Convergence of policy mirror ascent

The exponential update satisfies the KL telescoping inequality, at every s, h,

$$\begin{aligned}\langle Q_h^{\pi^t}, \pi_h^\star - \pi_h^t \rangle \le \langle Q_h^{\pi^t}, \pi_h^{t+1} - \pi_h^t \rangle \\ + \eta^{-1}\big(\mathrm{KL}(\pi_h^\star \| \pi_h^t) - \mathrm{KL}(\pi_h^\star \| \pi_h^{t+1})\big).\end{aligned}$$

The improvement inner product is nonnegative: multiplied by η, it equals $\mathrm{KL}(\pi_h^{t+1} \| \pi_h^t) + \mathrm{KL}(\pi_h^t \| \pi_h^{t+1})$. By performance difference and nonnegativity at every later stage,

$$\langle Q_h^{\pi^t}(s, \cdot), \pi_h^{t+1} - \pi_h^t \rangle \le V_h^{\pi^{t+1}}(s) - V_h^{\pi^t}(s).$$

Sum over t to telescope the values and KL terms. The resulting bound is $H + \log A/\eta$ for each s, h. Average under the fixed occupancy of $\pi^\star$ and sum over h to obtain

$$\sum_{t=1}^{T} (V_1^\star - V_1^{\pi^t})(s_1) \le H^2 + H \log A/\eta.$$

For fixed η, average suboptimality is $O(1/T)$. These guarantees assume exact policy evaluation; learning the unknown model also requires exploration. The 2020 comparison gives stochastic-reward policy-optimization regret of order $\widetilde{O}(\sqrt{\mathrm{poly}(H) S^2 AT})$, compared with $\widetilde{O}(\sqrt{\mathrm{poly}(H) SAT})$ for optimistic value iteration. For adversarial rewards, the policy-optimization bound is $O(T^{2/3})$, whereas upper occupancy bounds give $\widetilde{O}(\sqrt{\mathrm{poly}(H) S^2 AT})$.

15.3 Trajectory gradients and REINFORCE

For a differentiable policy with positive action probabilities, let $R(\tau) = \sum_h r_h(s_h, a_h)$. The likelihood-ratio identity gives

$$\nabla J(\theta) = \mathbb{E}_{\pi_\theta}\left[R(\tau)\sum_h \nabla \log \pi_{\theta,h}(a_h \mid s_h)\right] = \mathbb{E}_{\pi_\theta}\sum_h Q_h^{\pi_\theta}(s_h, a_h)\nabla \log \pi_{\theta,h}(a_h \mid s_h).$$

Proof. Differentiate $\sum_\tau \mathbb{P}_\theta(\tau)R(\tau)$ using $\nabla \mathbb{P}_\theta = \mathbb{P}_\theta \nabla \log \mathbb{P}_\theta$. Only the action probabilities depend on θ. Rewards before step h have zero contribution to its score, since the conditional mean of $\nabla \log \pi_{\theta,h}(a_h \mid s_h)$ is zero. Condition the remaining rewards on (s_h, a_h) to obtain the Q-value form. □

REINFORCE estimates the first expression with sampled trajectories and makes a gradient-ascent step. The full-trajectory estimator may have high variance. In direct tabular probabilities, its gradient is also scaled by state occupancy, whereas policy mirror ascent cancels that factor.

15.4 PPO and TRPO surrogates

For general parameters define a surrogate linear in the policy probabilities,

$$L_t(\theta) = \sum_{h,s} d_h^{\pi_{\theta^t}}(s)\langle Q_h^{\pi_{\theta^t}}(s, \cdot), \pi_{\theta,h}(\cdot \mid s) - \pi_{\theta^t,h}(\cdot \mid s)\rangle.$$

It need not be linear in θ. Let $D_t(\theta) = \sum_{h,s} d_h^{\pi_{\theta^t}}(s)\,\mathrm{KL}(\pi_{\theta,h} \| \pi_{\theta^t,h})$. The KL-penalized PPO formulation considered here maximizes $L_t(\theta) - D_t(\theta)/\eta$; its TRPO formulation maximizes L_t subject to $D_t \le \delta$. The former uses a penalty, the latter a trust-region constraint. In unrestricted tabular probabilities, the penalized update is policy mirror ascent. The stated formulations use new-to-old KL; the original practical formulations may use the reverse direction.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 15 (Spring 2020).

Lecture 16: Natural Policy Gradient and Linear Quadratic Regulators

16.1 Natural gradient and the Fisher matrix

For a smooth probability model p_θ, a local KL expansion is

$$\mathrm{KL}(p_\theta \| p_{\theta+\Delta}) = \tfrac{1}{2}\Delta^\top F_\theta \Delta + o(\|\Delta\|^2), \qquad F_\theta = \mathbb{E}_{p_\theta}[\nabla \log p_\theta \nabla \log p_\theta^\top].$$

The constant and first derivative vanish; differentiating the normalization $\int p_\theta = 1$ also gives $F_\theta = -\mathbb{E}\nabla^2 \log p_\theta$ under the usual differentiation conditions. Maximizing a linearized objective with this quadratic penalty gives the natural gradient update

$$\theta^{t+1} = \theta^t + \eta F_{\theta^t}^\dagger \nabla J(\theta^t).$$

Use the inverse when nonsingular, or the pseudoinverse on identifiable directions when the parameterization is redundant.

For trajectory probabilities, the score is a sum of action scores. Cross terms at distinct times vanish by conditioning on the history before the later action, whose score has conditional mean zero. Thus

$$F_\theta = \sum_h \mathbb{E}_{\pi_\theta}[\nabla \log \pi_{\theta,h}(a_h \mid s_h) \nabla \log \pi_{\theta,h}(a_h \mid s_h)^\top].$$

For tabular softmax policies $\pi_h(a \mid s) \propto e^{\theta_{h,s,a}}$, a natural-gradient direction is $Q_h^\pi(s,a)$ up to an action-independent baseline. Adding that direction to the logits yields the same exponential policy update as mirror ascent. The algorithms are not identical for arbitrary parameterizations.

16.2 Linear quadratic control

A linear quadratic regulator (LQR) has continuous state x_t and control u_t, dynamics $x_{t+1} = Ax_t + Bu_t$, and reward

$$r_t(x_t, u_t) = -x_t^\top Q x_t - u_t^\top R u_t, \qquad Q \succeq 0, \quad R \succ 0.$$

Here process noise is zero. Continuous spaces make tabular methods impractical, but the linear dynamics and quadratic objective give structure.

Example: For helicopter height h_t, velocity v_t, acceleration a_t, time step τ, and gravity g,

$$\begin{pmatrix} h_{t+1} \\ v_{t+1} \end{pmatrix} = \begin{pmatrix} 1 & \tau \\ 0 & 1 \end{pmatrix} \begin{pmatrix} h_t \\ v_t \end{pmatrix} + \begin{pmatrix} \tau^2/2 \\ \tau \end{pmatrix} (a_t - g).$$

To penalize $(h_t - h^\star)^2 + a_t^2$, augment the state by a constant: $x_t = (h_t, v_t, 1)^\top$, $u_t = a_t$. Then

$$A = \begin{pmatrix} 1 & \tau & -\tau^2 g/2 \\ 0 & 1 & -\tau g \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} \tau^2/2 \\ \tau \\ 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 0 & -h^\star \\ 0 & 0 & 0 \\ -h^\star & 0 & (h^\star)^2 \end{pmatrix}, \quad R = 1.$$

16.3 Riccati recursion

Theorem 16.1 [Finite-horizon LQR]. *With zero terminal reward, the optimal value and control have the form $V_t^\star(x) = -x^\top M_t x$ and $u_t^\star = -K_t x$, where*

$$\begin{aligned} M_{H+1} &= 0, \\ K_t &= (R + B^\top M_{t+1} B)^{-1} B^\top M_{t+1} A, \\ M_t &= Q + K_t^\top R K_t + (A - BK_t)^\top M_{t+1} (A - BK_t). \end{aligned}$$

Proof. Given the quadratic continuation value, the Bellman objective is $-x^\top Q x - u^\top R u - (Ax + Bu)^\top M_{t+1}(Ax + Bu)$. Its derivative in u gives the displayed K_t; $R > 0$ ensures a unique maximizer. Substituting it gives $M_t \succeq 0$. Backward induction completes the argument. □

Under suitable stabilizability and detectability conditions, the infinite-horizon recursion converges to the algebraic Riccati equation

$$M = Q + A^\top M A - A^\top M B (R + B^\top M B)^{-1} B^\top M A.$$

For known dynamics one can solve the recursion or optimize policy parameters. Unknown dynamics invite model estimation or direct value/policy learning.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 16 (Spring 2020).

Lecture 17: Function Approximation and Linear MDPs

17.1 Why function approximation is needed

Tabular sample bounds depend on the number of states. In images or robotics, the state space can be enormous or continuous, and observed states rarely repeat exactly. Function approximation shares information across states by restricting values or policies to a parameterized family. Actor–critic methods approximate both. The challenges are generalization, misspecification when the desired function is outside the family, and exploration. Begin with known features and a well-specified model.

17.2 Linear MDPs

Given $\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$, approximate $Q_h(s,a) = \phi(s,a)^\top w_h$. A linear MDP is a stronger assumption: for each stage,

$$r_h(s,a) = \phi(s,a)^\top \theta_h, \qquad P_h(\cdot \mid s,a) = \langle \phi(s,a), \mu_h(\cdot) \rangle,$$

where μ_h is a vector of signed measures and the combination is a probability measure. This implies

$$Q_h^\pi(s,a) = \phi(s,a)^\top w_h^\pi, \qquad w_h^\pi = \theta_h + \int V_{h+1}^\pi(s')\, d\mu_h(s')$$

for every policy, directly from the Bellman equation.

Example: One-hot features $\phi(s,a) = e_{(s,a)}$ recover tabular MDPs. Alternatively, simplex-valued features can mix d base transition distributions and reward coefficients, allowing infinitely many states. LQR is not generally a finite-dimensional linear MDP in this sense, despite having linear state dynamics.

Validity requires $\phi(s,a)^\top \mu_h(\mathcal{S}) = 1$ and $\phi(s,a)^\top \mu_h(B) \geq 0$ for every measurable B. Thus all feature vectors lie on an affine hyperplane: an entire Euclidean ball centered at zero cannot be the feature image of such an MDP.

17.3 Least-squares backups

For stage-h samples (s_i, a_i, r_i, s_i'), fit the Bellman targets:

$$\widehat{w}_h = \arg\min_w \sum_i \left(w^\top \phi(s_i,a_i) - r_i - \widehat{V}_{h+1}(s_i') \right)^2.$$

Writing $\phi_i = \phi(s_i, a_i)$, the normal equations give

$$\widehat{w}_h = \Lambda_h^{-1} \sum_i \phi_i (r_i + \widehat{V}_{h+1}(s_i')), \qquad \Lambda_h = \sum_i \phi_i \phi_i^\top .$$

Adding λI corresponds to ridge regularization and ensures invertibility. These backward regression steps define LEAST-SQUARES VALUE ITERATION (LSVI). Simulator access raises a design question: which state-action pairs should be queried? A small core set whose features span the useful directions can replace an exhaustive table.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 17 (Spring 2020).

Lecture 18: Least-Squares Value Iteration

18.1 Core sets

A core set is $\mathcal{K} = \{(s_i, a_i)\}_{i=1}^{d}$ with feature matrix $\Psi = [\phi(s_1, a_1), \dots, \phi(s_d, a_d)]$ invertible and

$$\|\Psi^{-1}\phi(s, a)\|_1 \le L \quad \text{for every } s, a.$$

Every feature is a linear combination of core features with total absolute coefficient at most L. For simplex features containing the vertices, choose those vertices: $\Psi = I$ and $L = 1$.

Algorithm 18.1: LSVI with a core set

For each h and $(s, a) \in \mathcal{K}$, query n independent successors
Set $\widehat{V}_{H+1} = 0$
for $h = H, H-1, \dots, 1$ **do**
    Fit $\widehat{w}_h$ by least squares to $r_h(s, a) + \widehat{V}_{h+1}(s')$
    $\widehat{Q}_h(s, a) \leftarrow \phi(s, a)^\top \widehat{w}_h$
    $\widehat{V}_h(s) \leftarrow [\max_a \widehat{Q}_h(s, a)]_{[0, H-h+1]}$
    $\widehat{\pi}_h(s) \in \arg\max_a \widehat{Q}_h(s, a)$

Here $[x]_{[l,u]}$ clips to $[l, u]$. Bounded continuation targets make the concentration argument valid even if linear extrapolation leaves the value range.

18.2 Sample guarantee

Theorem 18.1 [Core-set LSVI]. *For a linear MDP, $n = \widetilde{O}(L^2H^4/\epsilon^2)$ queries per core pair and stage suffice for an ϵ-optimal policy with high probability. Total query cost is $\widetilde{O}(L^2H^5d/\epsilon^2)$, independent of S.*

Proof. There are n copies of each core feature, so $\Lambda_h = n\Psi\Psi^\top$. The target mean $r_h + P_h\widehat{V}_{h+1}$ is linear in the features. Let ε_i be the difference between the empirical and true mean continuation value at core pair i. Since later-stage data are independent of stage-h data, Hoeffding and a union bound give $\max_i |\varepsilon_i| \le CH\sqrt{\iota/n}$, with $\iota = \log(Hd/\delta)$. The fitted backup error at an arbitrary pair is

$$|\widehat{Q}_h - (r_h + P_h\widehat{V}_{h+1})|(s, a) = |\phi(s, a)^\top\Psi^{-\top}\varepsilon| \le L\|\varepsilon\|_\infty =: e.$$

True Bellman operators are nonexpansive and clipping cannot increase error against a valid value. Backward induction gives $\|V_h^\star - \widehat{V}_h\|_\infty \le (H - h + 1)e$. For the greedy policy, the same

recursion compares $\widehat{V}_h$ with $V_h^{\widehat{\pi}}$, giving another $(H-h+1)e$. Thus policy loss is at most $2He \le CLH^2\sqrt{\iota/n}$. □

18.3 The fitted empirical operator

The unregularized least-squares backup can be written

$$\widehat{P}_h(\cdot \mid s,a) = \sum_j \phi(s,a)^\top \Lambda_h^{-1} \phi_j \, \delta_{s'_j}(\cdot).$$

Its coefficients may be negative, so it need not be a transition probability distribution. The guarantee uses controlled backup error and propagation through the true P_h, rather than treating this signed operator as a stochastic model.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 18 (Spring 2020).

Lecture 19: LSVI with UCB and Fitted Q-Iteration

19.1 Exploration with linear features

Without simulator access, a core set cannot be queried freely. The learner must collect diverse features along reachable trajectories. For episode k, write $\phi_h^\tau = \phi(s_h^\tau, a_h^\tau)$ and form

$$\begin{aligned}
\Lambda_h^k &= \lambda I + \sum_{\tau<k} \phi_h^\tau (\phi_h^\tau)^\top, \\
w_h^k &= (\Lambda_h^k)^{-1} \sum_{\tau<k} \phi_h^\tau \big(r_h(s_h^\tau, a_h^\tau) + \max_a Q_{h+1}^k(s_{h+1}^\tau, a)\big), \\
Q_h^k(s,a) &= \min\left\{H, \phi(s,a)^\top w_h^k + \beta\sqrt{\phi(s,a)^\top (\Lambda_h^k)^{-1}\phi(s,a)}\right\}.
\end{aligned}$$

Set $Q_{H+1}^k = 0$, compute these quantities backward, then act greedily for the episode. This is LSVI-UCB. The bonus measures uncertainty in the query feature direction. For one-hot features, its squared norm is $1/(\lambda + N_h^k(s,a))$, recovering the tabular visit-count bonus. Under bounded-feature and parameter normalizations, the stated regret is $\widetilde{O}(\sqrt{d^3H^3T})$, with suitable β and $T = KH$.

19.2 General function classes

Let $\mathcal{F}_h$ be candidate Q-functions and define

$$(T_h f)(s,a) = r_h(s,a) + \mathbb{E}_{s'\sim P_h(\cdot|s,a)} \max_{a'} f(s',a').$$

The central assumptions are realizability, $Q_h^\star \in \mathcal{F}_h$, and Bellman completeness, $T_h\mathcal{F}_{h+1} \subseteq \mathcal{F}_h$. With terminal function zero, completeness implies realizability by backward induction.

At each stage, suppose samples have $(s,a) \sim \mu_h$ and $s' \sim P_h(\cdot \mid s,a)$, with coverage

$$d_h^\pi(s,a) \le C\mu_h(s,a) \quad \text{for every policy } \pi.$$

Here d_h^π is the state-action occupancy. A simulator can provide uniform tabular samples, giving $C = SA$; small C is a substantial assumption in large spaces.

19.3 Fitted Q-iteration

For n samples at stage h, define

$$L_{D,h}(f, g) = \frac{1}{n} \sum_{(s,a,r,s') \in D_h} \big(f(s,a) - r - \max_{a'} g(s', a')\big)^2.$$

Initialize $\widehat{f}_{H+1} = 0$, fit $\widehat{f}_h \in \arg\min_{f \in \mathcal{F}_h} L_{D,h}(f, \widehat{f}_{h+1})$ backward, and return a greedy policy. Linear regression gives LSVI as a special case; the general regression oracle may be computationally difficult.

The population loss is

$$\mathbb{E} L_{D,h}(f, g) = \|f - T_h g\|_{2,\mu_h}^2 + \mathbb{E}_{\mu_h} \operatorname{Var}\big(\max_{a'} g(s', a') \mid s, a\big).$$

Expand around the conditional mean; the cross term vanishes. The additional variance depends on g but not f, so it does not change the minimizer for fixed g. Completeness makes the population backup attainable, and coverage transfers its correctness to reachable state-action pairs.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 19 (Spring 2020).

Lecture 20: FQI Analysis and Bellman Rank

20.1 Finite-sample FQI

Assume finite classes, $0 \le f_h \le B$ with $B \ge H$, terminal value zero, Bellman completeness, and coverage $d_h^\pi \le C\mu_h$. Uniform Hoeffding concentration over adjacent function pairs gives

$$|L_{D,h}(f,g) - \mathbb{E}L_{D,h}(f,g)| \le cB^2\sqrt{\frac{\log(H|\mathcal{F}|^2/\delta)}{n}} =: u, \quad |\mathcal{F}| = \max_h |\mathcal{F}_h|.$$

Compare the empirical minimizer $\widehat{f}_h$ with $T_h\widehat{f}_{h+1} \in \mathcal{F}_h$. The variance terms in their population losses cancel, yielding $\|\widehat{f}_h - T_h\widehat{f}_{h+1}\|_{2,\mu_h}^2 \le 2u$.

20.2 Propagating Bellman errors

For any candidate sequence f and any policy π,

$$\|Q_h^\star - f_h\|_{2,d_h^\pi} \le \sqrt{C}\sum_{t=h}^{H} \|f_t - T_t f_{t+1}\|_{2,\mu_t}.$$

Proof. Insert $T_h f_{h+1}$ and use the triangle inequality. Coverage bounds the local Bellman error. Jensen's inequality bounds the continuation difference by the next-stage Q-error under a policy selecting an action with largest absolute Q-error. Apply induction, using coverage for every such policy. The optimal Bellman operator is nonlinear; the argument uses its max inequality rather than distributing it over a difference. □

For the greedy policy π_f, the telescoping identity based on $V^\star$ expresses its loss as the sum along its trajectory of $V_h^\star(s_h) - Q_h^\star(s_h, \pi_{f,h}(s_h))$. Insert f_h at the optimal and greedy actions. Greediness makes the intermediate term nonpositive, and the preceding L_2 bound gives

$$V_1^\star - V_1^{\pi_f} \le 2H\sqrt{C}\sum_h \|f_h - T_h f_{h+1}\|_{2,\mu_h}.$$

Thus the elementary analysis gives

$$V_1^\star - V_1^{\widehat{\pi}} = O\left(H^2\sqrt{C}\,B\left(\frac{\log(H|\mathcal{F}|^2/\delta)}{n}\right)^{1/4}\right).$$

It suffices to take $n = O(H^8C^2B^4\log(H|\mathcal{F}|^2/\delta)/\epsilon^4)$ samples per stage. A sharper squared-loss

analysis improves this to $O(H^4 C B^2 \log(H|\mathcal{F}|^2/\delta)/\epsilon^2)$ per stage; total samples are Hn when each stage uses its own data. Finite discretizations explain why $\log|\mathcal{F}|$ is useful: it scales like $\widetilde{O}(SA)$ for tables and $\widetilde{O}(d)$ for bounded linear parameters.

20.3 Exploration and Bellman rank

Representative data may be difficult to obtain or may require large C. To describe exploration without assuming such a data set, let $\mathcal{F} = \mathcal{F}_1 \times \cdots \times \mathcal{F}_H$ and π_f be greedy for f. The AVERAGE BELLMAN ERROR under roll-in policy π is

$$\mathcal{E}_h(f, \pi) = \mathbb{E}_{s_h \sim d_h^\pi,\, a_h \sim \pi_{f,h}}[f_h(s_h, a_h) - (T_h f_{h+1})(s_h, a_h)].$$

Roll in with π for the first $h-1$ steps and use the action of π_f at step h. For finite $\mathcal{F}$, form the matrix with entries $E_h(f, f') = \mathcal{E}_h(f, \pi_{f'})$. The algebraic BELLMAN RANK is $\max_h \operatorname{rank}(E_h)$. Low-rank factorizations, together with the required boundedness conditions, enable sample-efficient exploration even when the state space is large; computational efficiency is a separate question.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, Lecture 20 (Spring 2020).

Lecture 21: Matrix Games and Markov Games

21.1 Matrix games

In a two-player zero-sum matrix game, the maximizing row player chooses a and the minimizing column player chooses b, giving payoff $R(a,b)$ to the first and $-R(a,b)$ to the second. Rock–paper–scissors has payoff matrix

$$R = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

There is no pure equilibrium: $\max_a \min_b R(a,b) = -1$ while $\min_b \max_a R(a,b) = 1$. A mixed strategy is a distribution $\mu \in \Delta_{\mathcal{A}}$ or $\nu \in \Delta_{\mathcal{B}}$, with expected payoff $\mu^\top R \nu$.

Theorem 21.1 [Minimax]. *For continuous f on compact convex domains, concave in the maximizing variable and convex in the minimizing variable, $\max_\mu \min_\nu f(\mu,\nu) = \min_\nu \max_\mu f(\mu,\nu)$. In a finite matrix game, optimal mixed strategies exist and satisfy*

$$\mu^\top R \nu^\star \le (\mu^\star)^\top R \nu^\star \le (\mu^\star)^\top R \nu \quad \text{for every } \mu, \nu.$$

Neither player benefits by deviating alone. Uniform play is an equilibrium in rock–paper–scissors. Projected gradient ascent for the maximizer and descent for the minimizer, or mirror updates, are natural optimization approaches; guarantees depend on the averaging and update scheme.

21.2 Zero-sum Markov games

A Markov game $\mathrm{MG}(\mathcal{S}, \mathcal{A}, \mathcal{B}, P, r, H)$ adds states and transitions $P_h(s' \mid s, a, b)$. At each step the players independently sample $a_h \sim \mu_h(\cdot \mid s_h)$ and $b_h \sim \nu_h(\cdot \mid s_h)$, receive opposite rewards, and transition. For fixed Markov policies,

$$\begin{aligned} Q_h^{\mu,\nu}(s,a,b) &= r_h(s,a,b) + (P_h V_{h+1}^{\mu,\nu})(s,a,b), \\ V_h^{\mu,\nu}(s) &= \sum_{a,b} \mu_h(a \mid s) Q_h^{\mu,\nu}(s,a,b) \nu_h(b \mid s). \end{aligned}$$

The terminal value is zero. Fixing one Markov policy leaves an MDP for the other player, so a Markov best response is sufficient and can be optimal at all states simultaneously. Define $V_h^{\mu,\dagger} = \min_\nu V_h^{\mu,\nu}$ and $V_h^{\dagger,\nu} = \max_\mu V_h^{\mu,\nu}$.

21.3 Equilibria and planning

Backward induction solves a matrix game at every state:

$$Q_h^{\star\star} = r_h + P_h V_{h+1}^{\star\star}, \qquad V_h^{\star\star}(s) = \max_{\mu_h} \min_{\nu_h} \mu_h^\top Q_h^{\star\star}(s) \nu_h.$$

Stagewise minimax gives Markov equilibrium policies $\mu^\star, \nu^\star$ for which $\max_\mu \min_\nu V_h^{\mu,\nu} = V_h^{\mu^\star,\nu^\star} = \min_\nu \max_\mu V_h^{\mu,\nu}$. This follows from the dynamic program even though whole-policy values are not generally concave-convex in the policy parameters.

The DUALITY GAP of (μ, ν) is

$$V_1^{\dagger,\nu}(s_1) - V_1^{\mu,\dagger}(s_1).$$

It is nonnegative, vanishes at equilibrium, and measures both players' vulnerability to best responses. For a fixed μ, best-response planning replaces the stagewise max-min by $\min_{\nu_h} \mu_h^\top Q_h \nu_h$. With a simulator, empirical transitions can be estimated at each (h, s, a, b). Interactive learning must additionally decide how upper and lower confidence values guide joint exploration.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, April 28, 2020.

Lecture 22: POMDPs and Predictive State Representations

22.1 Partial observability

A partially observable MDP has hidden states $\mathcal{S}$, actions $\mathcal{A}$, observations $\mathcal{O}$, transitions $\mathbb{T}(s' \mid s, a)$, emissions $\mathbb{O}(o \mid s)$, initial law μ_1, and horizon H. Consider stationary transitions and emissions. Rewards $r_h(o_h)$ are observed. Poker and games with fog of war motivate this distinction: observations need not reveal the underlying state.

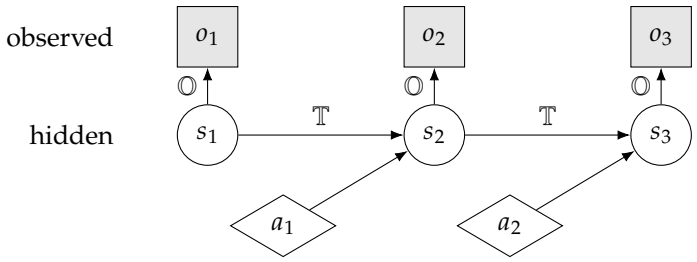


Figure 4: Hidden states evolve under the actions and emit observations. A policy may use the complete observed history.

Uncontrolled fully and partially observed processes are Markov chains and hidden Markov models; their controlled analogues are MDPs and POMDPs. The observation sequence is generally not Markov. A policy therefore maps $\tau_h = (o_1, a_1, \dots, a_{h-1}, o_h)$ to a distribution over actions.

22.2 History and belief-state planning

For consistency with a decision made after observing o_h, define remaining value from future rewards by

$$V_h^\pi(\tau_h) = \mathbb{E}_\pi\left[\sum_{t=h+1}^{H} r_t(o_t) \mid \tau_h\right].$$

Then $V_H = 0$ and

$$\begin{aligned} Q_h^\pi(\tau_h, a) &= \mathbb{E}\big[r_{h+1}(o_{h+1}) + V_{h+1}^\pi(\tau_h, a, o_{h+1}) \mid \tau_h, a\big], \\ V_h^\pi(\tau_h) &= \sum_a \pi_h(a \mid \tau_h) Q_h^\pi(\tau_h, a). \end{aligned}$$

The total objective includes the already observed first reward separately. Replacing the action average by a maximum gives the optimality equation. History has exponentially many possibilities in H.

The BELIEF STATE $b_h(s) = \mathbb{P}(s_h = s \mid \tau_h)$ is a sufficient statistic for predicting the future. After action a and observation o',

$$b_{h+1}(s') = \frac{\mathbb{O}(o' \mid s') \sum_s \mathbb{T}(s' \mid s, a) b_h(s)}{\sum_{\widetilde{s}} \mathbb{O}(o' \mid \widetilde{s}) \sum_s \mathbb{T}(\widetilde{s} \mid s, a) b_h(s)}.$$

Thus a POMDP is an MDP over beliefs in $\Delta_{\mathcal{S}}$. Bellman recursion uses the same observation expectation and this belief update in place of extending the history. The finite-horizon optimal value is piecewise linear in the belief, but the number of pieces may be exponential. The belief reduction does not make planning efficient in general. Unknown transitions and emissions add statistical difficulties.

22.3 Predictive state representations

Hidden states need not be directly interpretable or uniquely determined by observable behavior. A TEST is a future action-observation sequence $t = a^1 o^1 \cdots a^k o^k$. Let $p(t \mid h)$ be the probability of those observations after history h when the test's actions are externally fixed. The empty-history values $p(t)$ form the SYSTEM-DYNAMICS VECTOR. The SYSTEM-DYNAMICS MATRIX has histories as rows, tests as columns, and entry $p(t \mid h)$ for feasible histories. It is determined by joint test probabilities through $p(t \mid h) = p(ht)/p(h)$ under the specified action convention.

For a POMDP with S hidden states, $p(t \mid h) = b(h)^\top u_t$, where $u_t(s)$ is the test's success probability from hidden state s. Hence the dynamics matrix has rank at most S. Rank-three controlled processes already exist without any finite-state POMDP representation.

Choose k linearly independent test columns, the CORE TESTS $q_1, \ldots, q_k$, and define

$$\psi(h) = (p(q_1 \mid h), \ldots, p(q_k \mid h))^\top.$$

Every test has coefficients m_t independent of history such that $p(t \mid h) = m_t^\top \psi(h)$. This is a PREDICTIVE STATE REPRESENTATION (PSR), a sufficient statistic expressed entirely in observable predictions.

22.4 PSR update and composition

Let the ith row of M_{ao} be $m_{aoq_i}^\top$, and let m_{ao} predict the one-step test. Bayes' rule gives

$$\psi(hao) = \frac{M_{ao}\psi(h)}{m_{ao}^\top \psi(h)} \quad \text{when } p(ao \mid h) > 0.$$

Indeed, coordinate i is $p(q_i \mid hao) = p(aoq_i \mid h)/p(ao \mid h)$. Also $p(aot \mid h) = p(t \mid hao)p(ao \mid h)$ implies $m_{aot}^\top = m_t^\top M_{ao}$. Repeated composition therefore computes arbitrary test probabilities from $\psi(\varnothing)$, the vectors m_{ao}, and the matrices M_{ao}. These parameters specify observable system behavior without identifying a latent-state model.

Source. Chi Jin, *ELE524: Foundations of Reinforcement Learning*, April 30, 2020.